

Managing acute flank pain in adults

Version 65

For effective investigation and management of ureteric colic and its mimics

Do not use if urinary calculi known to be present or if patient is pregnant but seek senior urology advice

Disclaimer:
This is a clinical template; clinicians should always use judgment when managing individual patients

Reapproved by ED guidelines committee on 1st Apr 26
Review due Apr 31 Trust Ref: C71/2016

Patient details

Full name

DoB

S number

(use addressograph if available)

① Should NSAID be avoided?

☐ **YES**, at least one of the below

- Allergy to **ANY** NSAID
- Prior asthma worsening due to NSAIDs
- Known current peptic ulcer
- History or signs of heart failure or CAD
- Current antiplatelet or anticoagulant use
- Chronic kidney disease (CKD)
- Current illness that might cause AKI

☐ **NO**, none of the above

② Atypical features present?

☐ **YES**, at least one of the below

- Haemodynamic instability
- Expansile abdominal mass or known AAA
- Abdominal tenderness +/- peritonism
- New-onset abdominal distension
- Diarrhoea
- Non-colic pain character
- Temperature >37.9° C
- <1+ blood on urine dip
- Low Hb suggesting acute blood loss

☐ **NO**, none of the above

③ Alternative diagnoses

- Pyelonephritis
- Appendicitis
- Cholecystitis
- Diverticulitis
- Leaking AAA
- Pancreatitis
- Aortic dissection
- Basal pneumonia
- Bowel obstruction
- Testicular torsion
- Ovarian pathology
- Ectopic pregnancy
- Peptic ulcer
- Colitis
- MSK pain
- Hepatitis
- Nutcracker syndrome (NCS = left renal vein compression between aorta and SMA)
- Herpes zoster
- Pulmonary embolism
- Renal cell carcinoma
- Renal vein thrombosis
- Renal infarction
- Renal cysts
- 'Clot colic' caused by Loin pain haematuria syndrome (LPHS)
- Papillary infarction
- Ureteral transitional cell carcinoma (TCC)

④ Admission necessary?

☐ **YES**, as at least one of the below

- Temperature >37.9° C
- Other abnormal vital signs
- Significantly raised WBC count
- Acute renal impairment
- Stone in only functioning/transplanted kidney
- Bilateral obstructing stones
- Inadequate control of pain or vomiting

☐ **NO**, as none of the above

⑤ Stone discharge bundle

Arrange

- ☐ KUB X-ray before discharge (**ALWAYS** needed for follow-up planning)
- ☐ Follow-up by completing ICE referral (Service Refs > Urology > Stone MDT); ensure Ca²⁺ and urate results available

Prescribe (**NB:** If TrustMed closed, use FP10)

- ☐ Diclofenac TTO 50mg PR or PO (patient's choice but PR route works faster) TDS regularly for 5 days then PRN for 2 days
- ☐ Tamsulosin TTO 0.4mg PO OD for 4/52 **IF** stone in distal ureter **AND** >5mm **AND** no contraindications as per BNF

Advise

- ☐ Taking regular Paracetamol as well
- ☐ Return to ED if they develop uncontrolled pain, fever or rigors, or urinary retention
- ☐ Accessing the BAUS (British Association of Urological Surgeons) Patients' Page 'I think I might have kidney stones'

EITHER by searching online for 'baus patients kidney stones'

OR by pointing their phone to this QR code



- Ambulance and VAC/walk-in assessment nurses**
- Set initial DPS as per pain score and NEWS
 - Obtain IV access and 'Flank pain' order set bloods
 - Obtain midstream urine sample for
 - Dipstick urinalysis
 - Urine culture if dip suggestive of UTI
 - β-HCG in all women of childbearing potential
 - Request STAT assessment

- STAT clinician**
- Prescribe analgesia - Paracetamol IV STAT, oramorph PRN and, unless there are NSAID cautions (see box 1):
 - If aged less than 65 years: Diclofenac 100mg PR STAT
 - If aged 65 years or older: Ibuprofen 400mg PO instead
 - Add antiemetic as appropriate

- STAT or clerking clinician**
- If pain score still >4 after 15min: Morphine IV
 - Make a note of any atypical features (box 2) and alternative diagnoses (see box 3)
 - Clerking clinician to discuss with area lead
 - Adjust DPS according to response to analgesia and most likely clinical (differential) diagnosis

Is ureteric colic most likely cause of patient's symptoms?

N

Manage as appropriate

Request CTKUB

Atypical features (see box 2)?

N

eRefer to EDU on 'Flank pain pathway'

Await CTKUB report in the ED

Symptoms explained by non-stone CT findings?

N

Referral/disposition as appropriate

Signs of recent stone passage only, AND patient not taking indinavir?

Y

Stone reported OR (invisible) indinavir stone suspected?

Y

Admission needed (see box 4)?

N

Discharge unless still in pain or other reasons to admit
Make GP aware if Ca²⁺ or urate raised

Complete stone discharge bundle (see box 5)

- If any haematuria with persistent pain, acute renal impairment (or renal infarct or papillary necrosis) → discuss with renal 'registrar'
Exceptionally, if patient remains in severe & unremitting pain, ED senior (in liaison with renal team) might request CT urogram to look for renal infarct or papillary necrosis
- If visible (frank) haematuria → refer to urology OPD by sending completed [urology referral letter template](#) (on EDU guidance page) to uhl-tr-urologyadmin@nhs.net
- Otherwise discharge home and recommend the following to GP:
 - If persistent non-visible haematuria → to refer to urology OPD
 - If recurrent episodes of severe flank pain without obvious cause → to consider renal US with renal vessel Doppler to rule out NCS

Complete urology NC e-referral; under 'Reason for Referral' select 'Cross site Admission'

Seen by

Print name

Signature

Role

Date